IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

DISCRETE MATHEMATICS

Monday, 10 May 10:00 am

Time allowed: hours

There are THREE questions on this paper.

Answer questions.

Any special instructions for invigilators and information for candidates are on page 1.

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DISCRETE MATHEMATICS

1. a) For each of the following statements, state whether it is true or false and provide a supporting proof.

   i) $10n^2 + 2n^3 = O(n^3)$ [3]

   ii) $an + b2^n = O(2^n)$ for any $a \geq 0$, $b \geq 0$. [3]

   iii) $an^3 + bn^n = \Theta(n^n)$ for any $a \geq 0$, $b \geq 0$. [3]

   b) Give a tight bound for each of the following recurrence relations, or explain why it is not possible to do so.

   i) $T(n) = 7T(n/7) + n$. [2]

   ii) $T(n) = 9T(n/3) + \frac{n(n+1)}{2}$. [2]

   iii) $T(n) = 3T(n/2) + kn^2$, where $k > 0$. [2]

2. Consider the problem of searching for a value x in an unsorted array A, and returning either an index i so that $A[i] = x$, or -1 if x is not in A.

   a) What is the best achievable time complexity for unsorted search on a single threaded architecture? [2]

   b) Write pseudocode for a multi-threaded version using divide and conquer. Express your answer using the keywords `spawn` and `sync`. As a guideline, your answer should not need more than 15 lines of code. [4]

   c) Compute the work $T_1(n)$, span $T_\infty(n)$ and parallelism $T_1(n)/T_\infty(n)$ of your algorithm, carefully justifying your answers. [4]

3. Countup is a game of finding the shortest way to get to a target number n, starting at 0, with a fixed set of operations F. For example, if the target is $n = 6$ and the operations in F are A_1="add 1" and M_3="multiply by 3" then you could get to 6 by doing $A_1A_1A_1A_1A_1A_1 = 0+1+1+1+1+1+1$, but it would be shorter to do $A_1A_1M_3 = (0+1+1)\times 3$. Your goal is to efficiently cheat at this game using the power of algorithms.

Use the notation $\mathbb{N}^+ = \{1,2,3,\ldots\}$ for the set of positive integers. You may assume the following: $n \in \mathbb{N}^+$ is a positive integer; all operations $f \in F$ are of the form $f = A_i$ ="add i" or $f = M_j$ ="multiply by j", where $i, j \in \mathbb{N}^+$; in every case, $A_1 \in F$, guaranteeing that there is a solution of length at most n; the number of operations $|F|$ is fixed and does not depend on n.

a) A brute force solution to this problem is to iterate through every sequence of operations of length up to n, check if they match the target n, and return the shortest sequence. Compute the time complexity of this algorithm as a function of n. [4]

b) We can write a dynamic program to compute the length of the shortest solution as follows. Write f^{-1} to be the inverse function of f, so $A_i^{-1}(x) = x - i$, etc. Consider the problem of starting at n and finding the shortest sequence of inverse functions that arrives at 0. For the example of $n = 6$ and $F = \{A_1, M_3\}$ we get that $0 = A_1^{-1}(A_1^{-1}(M_3^{-1}(6)))$ which lets us reconstruct the solution $A_1A_1M_3$ to the original problem.

i) Write an equation for $C(n)$, the length of the shortest such sequence, in terms of $C(0), C(1), \ldots, C(n-1)$.

Hint: consider a series of choices taking you closer to 0. Make sure to check which choices are valid, and to consider the base case. [6]

ii) Without writing pseudocode, compute the time complexity of this algorithm. [4]

iii) Write pseudocode for this algorithm. As a guideline, it should not take more than 20 lines of code. You may write f^{-1} or `f.inverse` as the inverse of an operation. [6]

c) Find the set of all solutions with the shortest length in the case $n = 11$, $F = \{A_1, A_2, M_3, M_4\}$.

Hint: use the same basic structure of your dynamic program, but modify it to find all solutions; or alternatively use the brute-force algorithm. [5]